Deterministic Optimization

Discrete Optimization:

Linear Programming Relaxations

Shabbir Ahmed
Anderson-Interface Chair and Professor
School of Industrial and Systems Engineering

Ideal Formulations

Ideal Formulation

Learning Objectives

- Identify strong and ideal formulations of a mixed integer linear program

Formulations of a MILP

- The feasible region of a mixed integer linear program (MILP) is a set of the form
$$X = \{\mathbf{x} : \ \mathbf{x} \in P \cap (\mathbb{R}^{n-p} \times \mathbb{Z}^p)\}$$
where P is a polyhedral set.

- The same MILP can be formulated in different ways, e.g.
$$X = \{\mathbf{x} : \ \mathbf{x} \in Q \cap (\mathbb{R}^{n-p} \times \mathbb{Z}^p)\}$$
where Q is a different polyhedral set than P.

- We say the formulation using P is at least as strong as that using Q when $P \subseteq Q$

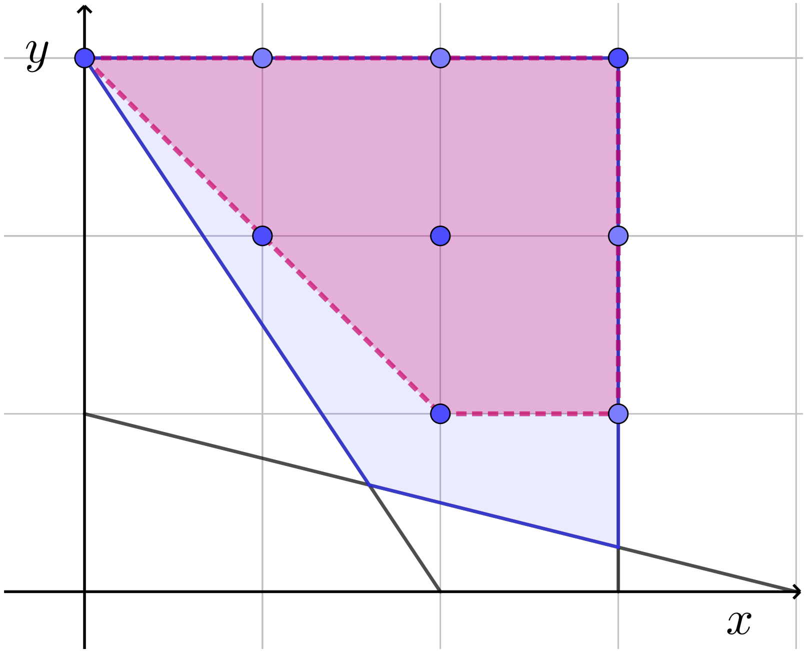
Example

Formulation 1:

$$X = \left\{ (x,y) : \begin{array}{l} 3x + 2y \geq 6 \\ x + 4y \geq 4 \\ 0 \leq x \leq 3,\ 0 \leq y \leq 3 \\ x,\ y \text{ integer} \end{array} \right\}$$

Formulation 2:

$$X = \left\{ (x,y) : \begin{array}{l} x + y \geq 3 \\ 0 \leq x \leq 3,\ 1 \leq y \leq 3 \\ x,\ y \text{ integer} \end{array} \right\}$$



Strong Formulations

- Stronger formulations (of an MILP) lead to stronger LP relaxations, and so better LP relaxation better bounds, and sometimes LP relaxation solutions that are feasible to the MILP

- The formulation of an MILP can be strengthened

  - by adding constraints (valid inequalities)
  - by tightening constraint coefficients
  - by introducing new variables and constraints (not covered)

Convex Hull / Ideal Formulation

- The convex hull of a set X, denoted by $\text{conv}(X)$ is the smallest convex set that contains X
- Fact: If X is the feasible region of a MILP, then $\text{conv}(X)$ is a polyhedral set.
- Fact: If X is the feasible region of a MILP, then all extreme points of $\text{conv}(X)$ are feasible solutions to the MILP
- Then
$$\min\{\mathbf{c}^\top \mathbf{x} : \mathbf{x} \in X\} \equiv \min\{\mathbf{c}^\top \mathbf{x} : \mathbf{x} \in \text{conv}(X)\}$$
- Thus MILP is equivalent to an LP with an ideal formulation!!

Strong Formulations

- If we had the convex hull or ideal formulation then we can solve an MILP by just solving an LP
- Getting the ideal formulation is as hard as (if not harder) to solve the MILP
- We try to get strong formulation that are closer to the ideal formulation so that the LP relaxation gives a good approximation of the MILP

Summary

- An ideal formulation of a MILP is one whose LP relaxation solves the MILP
- Ideal formulations are hard to obtain so we strive to obtain strong formulation that approximate the ideal formulation